Application 2:

选择定则的应用与直积

$(\psi_a, H'\psi_b)$ H' : perturbed Hamiltonian

$= \langle a|H'|b\rangle$

Scalar 不变，若变为0则为0

— selection rule

一. 处于微扰场的电磁相互作用

$$H = \frac{1}{2m}(\vec{p} - \frac{e}{c}\vec{A})^2 + V$$

固体中存在电磁场时电子的 H $(\nabla\cdot\vec{A}=0)$

$$H = \underbrace{\frac{p^2}{2m} + V(r)}_{H_0} \underbrace{- \frac{e}{mc}\vec{A}\cdot\vec{p} + \frac{e^2}{2mc^2}A^2}_{H'_{em}\approx -\frac{e}{mc}\vec{A}\cdot\vec{p}}$$

一般 H'_{em} 又可表为 $-\frac{e}{mc}\vec{A}\cdot\vec{p}$

(先记末尾有详细讨论)

本征函数 ψ_b 是 H_0 的某不可约表示的基函数的分量

$\psi_j^{(\Gamma_i)}$

二、基函数的正交性（对称性规则）

右上角 (i) → 不可约表示 Γ_i

右下角 α → 分量

$$\hat{P}_R \varphi_\alpha^{(i)} = \sum_{j=1}^{l_i} \varphi_j^{(i)} D^{(i)}(R)_{j\alpha}$$

$\hat{P}_R$：对称操作　　l_i 维不可约表示的基函数

independent of the Coordinate system

积分　　不变

$$(\phi_\alpha^{(i)}, \varphi_{\alpha'}^{(i')}) = (\hat{P}_R \phi_\alpha^{(i)}, \hat{P}_R \varphi_{\alpha'}^{(i')})$$

$$= \sum_{jj'} D^{(i)}(R)^*_{j\alpha} D^{(i')}_{j'\alpha'} (\phi_j^{(i)}, \varphi_{j'}^{(i')})$$

$$= \sum_{jj'} D^{(i)}(R)^*_{j\alpha} D^{(i')}_{j'\alpha'} (\phi_j^{(i)}, \varphi_{j'}^{(i')})$$

$$(\phi_\alpha^{(i)}, \varphi_{\alpha'}^{(i')}) = \frac{1}{h} \sum_R (\phi_\alpha^{(i)}, \varphi_{\alpha'}^{(i')})$$

$$= \frac{1}{h} \sum_R \sum_{jj'} D^{(i)*}_{j\alpha}(R) D^{(i')}_{j'\alpha'}(R) (\phi_j^{(i)}, \varphi_{j'}^{(i')})$$

$$= \frac{1}{l_i} \sum_{jj'} \delta_{ii'} \delta_{jj'} \delta_{\alpha\alpha'} (\phi_j^{(i)}, \varphi_{j'}^{(i')})$$

$$= \frac{1}{l_i} \delta_{ii'} \delta_{\alpha\alpha'} \sum_{j=1}^{l_i} (\phi_j^{(i)}, \varphi_j^{(i)})$$

$= \delta_{ii'} \delta_{\alpha\alpha'}$

If a symmetry operator is invariant under all of the symmetry operations of the group of Schrödinger's equation then it transforms like the identity operator

$$H_0 \psi_{\alpha'}^{(i')} = E_{\alpha'}^{(i')} \psi_{\alpha'}^{(i')}$$

$E_{\alpha'}^{(i')}$ is a number (or eigenvalue) which is independent of any coordinate system.

$\phi_\alpha^{(i)}, \psi_\alpha^{(i')}$

$\left(\phi_\alpha^{(i)}, H_0 \psi_\alpha^{(i')} \right) = 0$ eigenfunctions of H_0

unless $i = i'$ & $\alpha = \alpha'$

In the more case when we have a perturbation H', the perturbation need not have the full symmetry of H_0.

$H'\psi$ transforms differently from ψ

三. 直积群

e.g. $D_{3d} = D_3 \otimes \sigma_d$

$D_{3h} = D_3 \otimes \sigma_h$

$D_h = O \otimes C_i$

四. 不可约表示的直积

直积矩阵 $A \otimes B = C$

$A_{ij} B_{kl} = C_{ik,jl}$

2×2 , 2×2

$A_{2\times2} \otimes B_{2\times2}$

11 , 12 , 21 , 22

↓ ↓ ↓ ↓

1 2 3 4

matlab 排序方式

```
>> a=1:1:16

a =

     1     2     3     4     5     6     7     8     9    10    11    12    13    14    15    16

>> reshape(a,[2,2,2,2])

ans(:,:,1,1) =

     1     3
     2     4


ans(:,:,2,1) =

     5     7
     6     8


ans(:,:,1,2) =

     9    11
    10    12


ans(:,:,2,2) =

    13    15
    14    16
```

群表示的直积构成了直积群的表示。

两个群的不可约表示的直积构成直积群的不可约表示。

五. 直积的特征标

(a) $\chi^{(a\otimes b)}(A_k B_l) = \chi^{(a)}(A_k)\,\chi^{(b)}(B_l)$

$A_k \in G_A$

$B_l \in G_B$

(b) $\chi^{(b_1\otimes b_2)}(R) = \chi^{(b_1)}(R)\,\chi^{(b_2)}(R)$

证明：

已知 $D^{(a\otimes b)}(A_k B_l)_{ip,jq} = D^{(a)}(A_k)_{ij}\, D^{(b)}(B_l)_{pq}$

取对角元求和

$$\sum_{ip} D^{(a\otimes b)}(A_k B_l)_{ip,ip} = \sum_i D^{(a)}(A_k)_{ii} \sum_p D^{(b)}(B_l)_{pp}$$

$$\Rightarrow \chi^{(a\otimes b)}(A_k B_l) = \chi^{(a)}(A_k)\,\chi^{(b)}(B_l)$$

$$\chi^{(\lambda)}(R)\;\chi^{(\mu)}(R) = \sum_{\nu} a_{\lambda\mu\nu}\,\chi^{(\nu)}(R)$$

ν: label irreducible reps

可约表示特征标分解公式

$$\chi^{\text{reducible}}(C_k) = \sum_{P_j} a_j\,\chi^{(P_j)}(C_k)$$

$$a_j = \frac{1}{h}\sum_k N_k\,\chi^{(P_j)}(C_k)^*\,\chi^{\text{reducible}}(C_k)$$

h: group order; N_k: # of elements in class k

此处一下得

$$a_{\lambda\mu\nu} = \frac{1}{h}\sum_{C_\alpha} N_{C_\alpha}\,\chi^{(\nu)}(C_\alpha)^*\left[\chi^{(\lambda)}(C_\alpha)\,\chi^{\mu}(C_\alpha)\right]$$

C_4	E	C_2	C_4	C_4^3
A	1	1	1	1
B	1	1	-1	-1
E_1	1	-1	i	-i
E_2	1	-1	-i	i

$C_i = \{E, i\}$

$C_{4h} = C_4 \otimes i$

C_{4h}	E	C_2	C_4	C_4^3	i	$iC_2 = \sigma_h$	iC_4	iC_4^3
A_g	1	1	1	1	1	1	1	1
B_g	1	1	-1	-1	1	1	-1	-1
E_{1g}	1	-1	i	-i	1	-1	i	-i
E_{2g}	1	-1	-i	i	1	-1	-i	i
A_u								
B_u								
E_{1u}								
E_{2u}								

↓ ×(1)

↓ ×(-1)

gerade zahl 偶

ungerade 奇

非偶

C_i	E	i
	1	1
	1	-1

六、 群论在光谱选择定则

含波函数 $\psi_\alpha^{(i)}$ 和 $\psi_{\alpha'}^{(i')}$ 分别随 Γ_i 和 $\Gamma_{i'}$ 不可约表示的分量 α 和 α' 变换，并令 μ' 随表示 Γ_j 变换。如果直积 $\Gamma_j \otimes \Gamma_{i'}$ 与 Γ_i 正交，则矩阵元为零。

或者换个说法，如果 $\Gamma_i \otimes \Gamma_j \otimes \Gamma_{i'}$ 不包含完全对称表示 Γ_1，则矩阵元为零。

$$\mu' = \sum_{j,\beta} f_\beta^{(j)} \mu_\beta'^{(j)}$$

j : irreducible representations Γ_j of μ'

β : partners of Γ_j

$$\mu_\beta'^{(j)} \otimes \psi_\alpha^{(i)} \longrightarrow \Gamma_{j,\beta} \otimes \Gamma_{i\alpha}$$

Symmetry notation

$$\left(\psi_{\alpha'}^{(i')}, \mathcal{H}'\phi_{\alpha}^{(i)}\right) = 0$$

if $\psi_{\alpha'}^{(i')} \perp \{\mathcal{H}'\phi_{\alpha}^{(i)}\}$

or another interesting perspective:

$$\psi_{\alpha'}^{(i')} \otimes \mathcal{H}_{\beta}'^{(j)} \otimes \phi_{\alpha}^{(i)}$$

in order for the matrix element in $\left(\psi_{\alpha'}^{(i')}, \mathcal{H}'\phi_{\alpha}^{(i)}\right)$ to be nonzero, this triple direct product must contain a term that transforms as a scalar or a constant number, or according to the irreducible representation Γ_1.

两种视角的等价性：

$$\langle \psi_1 | H' | \psi_2 \rangle$$

1° $\Gamma^{(H')} \otimes \Gamma^{(2)}$ 直积表示分解

$$a_1 = \frac{1}{h} \sum_k N(k) \Gamma^{(1)*}(C_k)$$

$$(\Gamma^{(H')}(C_k) \otimes \Gamma^{(2)}(C_k))$$

如果 $a_1 \neq 0$，则 $\langle \psi_1 | H' | \psi_2 \rangle \neq 0$

2° 三直积表示向全对称表示的分解

设全对称表示为 Γ_0

$$a_0 = \frac{1}{h} \sum_k N_k \Gamma^{(1)*}(C_k)$$

$$[\Gamma^{(1)*}(C_k) \otimes \Gamma^{(H')}(C_k) \otimes \Gamma^{(2)}(C_k)]$$

due to the calculation of the matrix element

$$\langle \psi_1 | H' | \psi_2 \rangle = \int \psi_1^* H' \psi_2 \, d\vec{\tau}$$

$\because \Gamma^{(0)}(C_k) = 1$

$$a_0 = a_1$$

待处理的问题：

考虑在具有 O_h 对称性的系统中的电偶极跃迁

$$\mathcal{H}'_{em} = -\frac{e}{mc}\vec{A}\cdot\vec{p}$$

$\vec{A}$ 是外场，不参与系统的对称性变换

已 set up

常函数，fix 坐标系，→

fix 观测者

系统是电子系统

相对于坐标系的主动变换

$\mathcal{H}' = -\frac{e}{mc}\vec{A}\cdot\vec{p}$ 随 $\vec{p}$ 变换

$T_1^u(x,y,z)$

[$\vec{A}(\vec{r})$ 是空间依赖的，但在系统看来是常量

场，如果将环境+系统打包在一起，

$A(\vec{r})$ 将卷入与空间相关的变换，这也同

时说明了此处讨论的适应范围了

2020.4.17

类似统计中关于

系统与环境的讨论

Table 10.2. Character table for the cubic group O_h corresponding to the group of the wave vector at $k = 0$ for the three cubic space groups #221 (SC), #225 (FCC), and #229 (BCC)†

Simple cubic

	repr.	basis functions	E	$3C_4^2$	$6C_4$	$6C_2'$	$8C_3$	i	$3iC_4^2$	$6iC_4$	$6iC_2'$	$8iC_3$
A_{1g}	$\Gamma_1\,(\Gamma_1^+)$	1	1	1	1	1	1	1	1	1	1	1
A_{2g}	$\Gamma_2\,(\Gamma_2^+)$	$x^4(y^2-z^2)+y^4(z^2-x^2)+z^4(x^2-y^2)$	1	1	−1	−1	1	1	1	−1	−1	1
E_g	$\Gamma_{12}\,(\Gamma_{12}^+)$	x^2-y^2, $2z^2-x^2-y^2$	2	2	0	0	−1	2	2	0	0	−1
T_{1u}	$\Gamma_{15}(\Gamma_{15}^-)$	x, y, z	3	−1	1	−1	0	−3	1	−1	1	0
T_{2u}	$\Gamma_{25}(\Gamma_{25}^-)$	$z(x^2-y^2)\ldots$	3	−1	−1	1	0	−3	1	1	−1	0
A_{1u}	$\Gamma_1'\,(\Gamma_1^-)$	$xyz[x^4(y^2-z^2)+y^4(z^2-x^2)+z^4(x^2-y^2)]$	1	1	1	1	1	−1	−1	−1	−1	−1
A_{2u}	$\Gamma_2'(\Gamma_2^-)$	xyz	1	1	−1	−1	1	−1	−1	1	1	−1
E_u	$\Gamma_{12}'\,(\Gamma_{12}^-)$	$xyz(x^2-y^2)\ldots$	2	2	0	0	−1	−2	−2	0	0	1
T_{1g}	$\Gamma_{15}'\,(\Gamma_{15}^+)$	$xy(x^2-y^2)\ldots$	3	−1	1	−1	0	3	−1	1	−1	0
T_{2g}	$\Gamma_{25}'\,(\Gamma_{25}^+)$	xy, yz, zx	3	−1	−1	1	0	3	−1	−1	1	0

† The basis functions for Γ_{25}^- are $z(x^2-y^2)$, $x(y^2-z^2)$, $y(z^2-x^2)$, for Γ_{12}^- are $xyz(x^2-y^2)$, $xyz(3z^2-r^2)$ and for Γ_{15}^+ are $xy(x^2-y^2)$, $yz(y^2-z^2)$, $zx(z^2-x^2)$

假设初态为 T_{2g} (xy, yz, zx) 对称性的态

直积表示的特征标：

$\chi_{T_{1u}}\ \chi_{T_{2g}} = 9, 1, -1, -1, 0,$

$-9, -1, 1, 1, 0$

为5与5个特征标乘积相加。

$$a^{(T_{1u}\otimes T_{2g},\ A_{1g})} = \frac{1}{48}(1\times1\times9 + 3\times1\times1 + 6\times1\times(-1) + 6\times1\times(-1) + 8\times1\times0 + 1\times1\times(-1) + 3\times1\times(-1) + 6\times1\times1 + 6\times1\times1 + 8\times1\times0) = 0$$

$$a^{(T_{1u}\otimes T_{2g},\ A_{2g})} = 0$$

$$a^{(T_{1u}\otimes T_{2g},\ T_{1u})} = 1$$

.....

都是奇宇称.

$$T_{1u}\otimes T_{2g} = \frac{A_{2u} + E_u + T_{1u} + T_{2u}}{\text{(宇是奇宇称)}}$$

初态为偶宇称复合奇宇称，末态只能是奇宇称.

different initial states

$$T_{1u}\otimes E_g = T_{1u} + T_{2u} \quad \text{(奇宇称)}$$

$$T_{1u}\otimes T_{1u} = A_{1g} + E_g + T_{1g} + T_{2g} \quad \text{(偶宇称)}$$

3维 空间移动的表示是偶宇称 $\det = 1$，

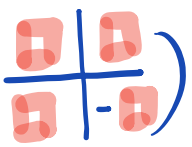
反演 $\det\begin{pmatrix} -1 & & \\ & -1 & \\ & & -1 \end{pmatrix} = -1$

O_h	E	$3C_4^2$	$6C_4$	$6C_2'$	$8C_3$	i	$3iC_4^2$	$6iC_4$	$6iC_2'$	$8iC_3$
A_1^g	1	1	1	1	1					
A_2^g	1	1	-1	-1	1					
E^g	2	2	0	0	-1			1× block		
T_1^g	3	-1	1	-1	0					
T_2^g	3	-1	-1	1	0					
(xy, yz, zx)										
A_1^u										
A_2^u										
E^u			1× block					(-1)× block		
T_1^u (x, y, z)										
T_2^u										

$\begin{pmatrix} 1 & & \\ & 1 & \\ & & -1 \end{pmatrix}$ 按宇称分块

施加压力：$O_h \rightarrow D_{4h}$

节 矢量：奇宇称奇函数

{e, ij $\begin{pmatrix} 1 & & \\ & 1 & \\ & & -1 \end{pmatrix}$ → 

Character table for point group D_{4h}

(x axis coincident with C'_2 axis)

D_{4h}	E	$2C_4$ (z)	C_2	$2C'_2$	$2C''_2$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$	linear functions, rotations	quadratic functions	cubic functions
A_{1g}	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1	-	x^2+y^2, z^2	-
A_{2g}	+1	+1	+1	-1	-1	+1	+1	+1	-1	-1	R_z	-	-
B_{1g}	+1	-1	+1	+1	-1	+1	-1	+1	+1	-1	-	x^2-y^2	-
B_{2g}	+1	-1	+1	-1	+1	+1	-1	+1	-1	+1	-	xy	-
E_g	+2	0	-2	0	0	+2	0	-2	0	0	(R_x, R_y)	(xz, yz)	-
A_{1u}	+1	+1	+1	+1	+1	-1	-1	-1	-1	-1	-	-	-
A_{2u}	+1	+1	+1	-1	-1	-1	-1	-1	+1	+1	z	-	$z^3, z(x^2+y^2)$
B_{1u}	+1	-1	+1	+1	-1	-1	+1	-1	-1	+1	-	-	xyz
B_{2u}	+1	-1	+1	-1	+1	-1	+1	-1	+1	-1	-	-	$z(x^2-y^2)$
E_u	+2	0	-2	0	0	-2	0	+2	0	0	(x, y)	-	(xz^2, yz^2) (xy^2, x^2y), (x^3, y^3)

$z \rightarrow A_{2u}$

$(x, y) \rightarrow E_u$

(x, y, z)

T_{1u} (in O_h) → $A_{2u} + E_u$ (in D_{4h})

e.g. xy : 两个极矢量乘积 (偶宇称)

(xy, x^2, y^2) $\qquad$ (x^2, y^2) $\quad xy$

T_{2g} (in O_h) $\rightarrow$ $E_g + B_{2g}$ (in D_{4h})

(in O_h)

$E_g \otimes T_{1u} = T_{1u} + T_{2u}$

$E_g \otimes (A_{2u} + E_u) = E_u + (A_{1u} + A_{2u} + B_{1u} + B_{2u})$

(in D_{4h}) $\qquad$ (D_{4h} 中所有奇宇称)

(*)

D_{4h} 是 O_h 的子群

$O_h \rightarrow D_{4h}$, 对称性降低,

选择定则的限制减弱, 允许更多的跃迁。

如果考虑 z 偏振, ~~E_u~~, 只剩下 A_{2u}

初态 $E_g \otimes A_{2u} = E_u$

(*) RHS 中 $A_{1u} + A_{2u} + B_{1u} + B_{2u}$ 也消失

初态 $E_u \otimes E_u = A_{1g} + A_{2g} + B_{1g} + B_{2g}$

(E_u: 面内偏振) $\qquad$ (奇 × 奇 → 偶)

A_{xu} (z)

E_u (x, y)

刻画 $\vec{r}$ 的不可约表示标记

$$H_{int} \approx -\frac{e}{mc}\vec{A}\cdot\vec{p} \quad (H')$$

$\vec{A}$ 代表电磁场矢势，其方向为光子的极化方向 $\vec{r}$（或等价偏极角省 $e\vec{r}$）是物质系统响应部分

光的极化在 X-Y 面内，则由 E_u 表示决定

沿 z 轴，则由 A_{xu} 决定

Selection rules：

$$\langle f|\vec{r}|i\rangle$$

这对应辐射的极化（polarized radiation）

物理结果表现为：系统对不同方向极化的电磁波表现出完全不同的吸收、反射或辐射能力。

向波 《Building blocks of quantum mechanics》

问1：双光子过程吗？

10.2.1 INTERACTION OF ATOMS WITH ELECTROMAGNETIC WAVE

The interaction of an electron with an applied electromagnetic field is governed by the gauge-invariant Hamiltonian,

$$H = \frac{1}{2\mu}(\mathbf{p} - e\mathbf{A})^2 = \frac{\mathbf{p}^2}{2\mu} - \frac{e}{2\mu}(\mathbf{p}\cdot\mathbf{A} + \mathbf{A}\cdot\mathbf{p}) + \frac{e^2}{2\mu}\mathbf{A}^2. \tag{10.50}$$

If we take the kinetic energy of the free particle, i.e. the first term on the right-hand side of the above equation, as the unperturbed Hamiltonian, the perturbation is simply given by

$$H_1 = -\frac{e}{2\mu}(\mathbf{p}\cdot\mathbf{A} + \mathbf{A}\cdot\mathbf{p}). \tag{10.51}$$

The $\mathbf{A}^2$-term does not interact with the particle and is omitted.

We assume $\mathbf{A}$ to be the gauge field generated by a monochromatic field of plane wave

$$\mathbf{A} = A_0\hat{\boldsymbol{\varepsilon}}\cos(\mathbf{k}\cdot\mathbf{r} - \omega_0 t). \tag{10.52}$$

A_0 is the amplitude and $\hat{\boldsymbol{\varepsilon}}$ is a unit polarization vector. The corresponding electric field is

$$\mathbf{E} = -\frac{\partial\mathbf{A}}{\partial t} = A_0\omega_0\hat{\boldsymbol{\varepsilon}}\sin(\mathbf{k}\cdot\mathbf{r} - \omega_0 t). \tag{10.53}$$

As the electromagnetic wave is a transverse wave, $\hat{\boldsymbol{\varepsilon}}$ is perpendicular to the propagation direction, i.e. $\hat{\boldsymbol{\varepsilon}}\cdot\mathbf{k} = 0$, so $\nabla\cdot\mathbf{A} = \mathbf{0}$ and H_1 can be simplified as

$$H_1 = -\frac{e}{\mu}\mathbf{A}\cdot\mathbf{p} = -\frac{e}{\mu}A_0(\hat{\boldsymbol{\varepsilon}}\cdot\mathbf{p})\cos(\mathbf{k}\cdot\mathbf{r} - \omega_0 t). \tag{10.54}$$

In a hydrogenic system, the energy of the radiation field, $\hbar\omega_0$, must be of the order of atomic level spacing,

$$\hbar\omega_0 \sim \frac{Ze^2}{4\pi\varepsilon_0 R_{\text{atom}}}. \tag{10.55}$$

The corresponding wave length is

$$\lambda = \frac{2\pi c}{\omega_0} = \frac{2\pi R_{\text{atom}}}{Z\alpha} \approx \frac{274\pi R_{\text{atom}}}{Z}, \tag{10.56}$$

A1：$\vec{m}$ 为0，

$$\vec{A}(\vec{r},t) \propto a e^{i\vec{k}\cdot\vec{r}} + a^\dagger e^{-i\vec{k}\cdot\vec{r}}$$

$$H_{A^2} \propto \vec{A}\cdot\vec{A} \quad (aa,\ a^\dagger a^\dagger,\ aa^\dagger,\ a^\dagger a)$$

where

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 c\hbar} \approx \frac{1}{137} \tag{10.57}$$

is the fine structure constant. For a light atom (small Z), the wave length is much larger than the atom radius R_{atom}

$$\lambda \gg R_{\text{atom}}. \tag{10.58}$$

Therefore, we can ignore the $\mathbf{r}$-dependence of the electromagnetic wave and approximate H_1 by

$$H_1 \approx -\frac{e}{\mu} A_0 (\hat{\varepsilon} \cdot \mathbf{p}) \cos(\omega_0 t). \tag{10.59}$$

This is just the monochromatic perturbation defined in Eq. (10.42) with

$$H' = -\frac{e}{\mu} A_0 (\hat{\varepsilon} \cdot \mathbf{p}). \tag{10.60}$$

The matrix element of H' is

$$\langle n | H' | k \rangle = -\frac{e}{\mu} A_0 \langle n | \hat{\varepsilon} \cdot \mathbf{p} | k \rangle. \tag{10.61}$$

Using the equation

$$[\mathbf{r}, H_0] = \frac{i\hbar}{\mu} \mathbf{p}, \tag{10.62}$$

we can express this matrix element as

$$\langle n | H' | k \rangle = \frac{e}{i\hbar} A_0 (E_n - E_k) \hat{\varepsilon} \cdot \langle n | \mathbf{r} | k \rangle = -ieA_0 \omega_{nk} \langle n | \hat{\varepsilon} \cdot \mathbf{r} | k \rangle. \tag{10.63}$$

Since $e\mathbf{r}$ is the dipole operator of electron, one can regard H' as an effective interaction induced by the electric dipole interaction. Therefore, the corresponding transition rate in the long time limit, according to Eq. (10.49), is

$$\begin{aligned} w_{nk}(t) &= \frac{\pi e^2 A_0^2 \omega_{nk}^2}{2\hbar^2} |\langle n | \hat{\varepsilon} \cdot \mathbf{r} | k \rangle|^2 [\delta(\omega_{nk} - \omega_0) + \delta(\omega_{nk} + \omega_0)] \\ &= \frac{\pi e^2 A_0^2 \omega_0^2}{2\hbar^2} |\langle n | \hat{\varepsilon} \cdot \mathbf{r} | k \rangle|^2 [\delta(\omega_{nk} - \omega_0) + \delta(\omega_{nk} + \omega_0)]. \end{aligned} \tag{10.64}$$

$\vec{A}\cdot\vec{p}$, $q\vec{r}$

在选择定则的判断上相同

(宇称关系)

从物理上看：

$$H = \frac{p^2}{2m}$$

$$[\hat{r}, H] = \frac{1}{2m}[\hat{r}, \hat{p}^2]$$

$$= \frac{1}{2m}[\hat{r}, \hat{p}]\hat{p} + \hat{p}[\hat{r}, \hat{p}]$$

$$= \frac{2i\hbar}{2m}\hat{p} = \frac{i\hbar}{m}\hat{p}$$

$$\hat{p} = \frac{m}{i\hbar}[\hat{r}, H]$$

$$\langle f | \frac{e}{mc}\vec{p}\cdot\vec{A} | i \rangle \sim \langle f | \vec{p}\cdot\vec{A} | i \rangle$$

$$\sim \langle f | [\hat{r}, H] | i \rangle$$

$$= (E_i - E_f)\langle f | \hat{r} | i \rangle$$

$$\propto \langle f | \hat{r} | i \rangle$$

Q2: How to identify the symmetry of $\hat{A}$?

A2: $\hat{A}$不变（与 Schrödinger 方程的 Hilbert 空间无关）

电路元是是外加的 bath or reservior

Q3: 什么叫矢量？

A3: 跟页一样多。

Q4: 什么叫拥有 XX 的称性？

A4: 跟 XX 对称群不可约表示的基函数一样多